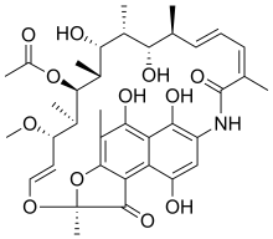
고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

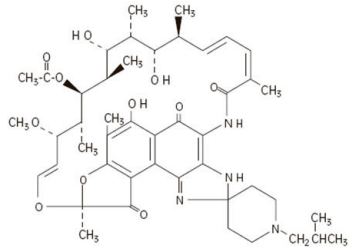
KU-The Future
110 years and beyond

Microbiology



Fall semester, 2026





❖ Nucleic acid synthesis inhibitors

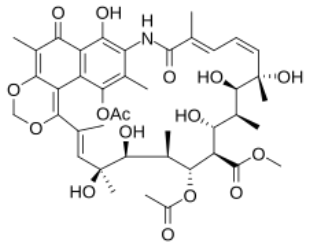
▶ Rifamycins

· a class of ansamycins: streptovaricins and rifamycins

· Natural products:
  – be produced from a species of soil bacteria (actinomycetes)
    : *Streptomyces mediterranei* (1957) → *Nocardia mediterranei* (1969) → *Amycolatopsis mediterranei* (1986) → *Amycolatopsis rifamycinica* (2004)
  – rifamycin A, B, C, D, E, S, and SV

· Semi-synthetic products: rifampin, rifabutin, etc.

· be easily absorbed from the digestive tract
  – except when taken directly after a meal



Streptovaricin A

· be bactericidal and have a wide spectrum of activity (many G (+) and some G (–) bacteria)

  → but, be generally used for tubeculosis (be active against most mycobacteria)

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❖ Nucleic acid synthesis inhibitors

▶ Rifamycins

· Adverse effects:

- liver damage when excessive doses are given
- red man syndrome: accumulation of colored metabolic products in the body (bright orange or red urine, saliva, tear, and skin)

· Mechanism of action:

- high affinity to bacterial RNA polymerase

→ inhibiting bacterial (DNA-dependent) RNA synthesis

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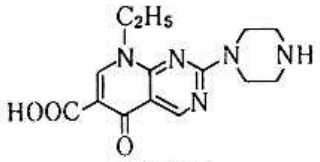
❖ Nucleic acid synthesis inhibitors

▶ Quinolones

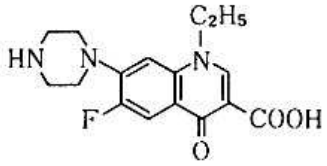
- synthetic bactericidal analogs of nalidixic acid
- wide/broad-spectrum antibiotics: effective against G(+) and G(−) bacteria (from the 2nd generation)
- classification:
  - DNA topoisomerase II inhibitors:
    - 1st generation: nalidixic acid and its derivatives (e.g., pipemidic acid)
      - nalidixic acid: the predecessor of all the quinolones
      - mainly effective against G(−)
    - 2nd generation: fluoroquinolones (norfloxacin, ciprofloxacin, ofloxacin, etc.) / effective against both G(+) and G(−) bacteria
  - DNA topoisomerase II and IV inhibitors (dual action):
    - 3rd generation: fluoroquinolones (tosufloxacin, sparfloxacin, levofloxacin, etc.) / (unlike the 1st and 2nd) active against genera *Staphylococci* and *Streptococci*
    - 4th generation: fluoroquinolones (moxifloxacin, trovafloxacin, etc.)
- Adverse effects: excitotoxic type effects and oxidative stress



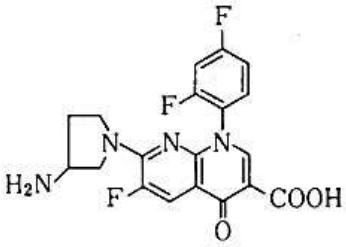
날리딕스산



피페미드산



노르플록사신

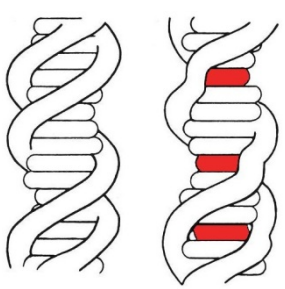


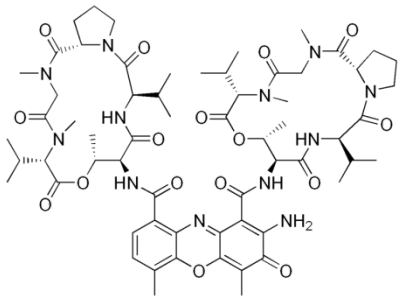
토수플록사신



스파르플록사신

❖ Nucleic acid synthesis inhibitors





▶ Antitumor agents

: intercalators and chemical modifiers

1. Intercalators:

· small molecules that insert themselves into the structure of DNA (intercalation)

· actinomycin D:

– be the first antibiotic shown to have anti-cancer activity
– first isolated by Selman Waksman from genus *Streptomyces*
– mechanism: intercalating into DNA (dsDNA; not ssDNA or RNA)
  · DNA replication inhibition
  · transcription inhibition
– spectrum of activity: a variety of cancers and dsDNA viruses
– side effects: fatigue, loss of appetite, diarrhea, hair loss, bone marrow suppression, etc.

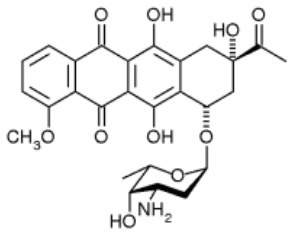
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❖ Nucleic acid synthesis inhibitors

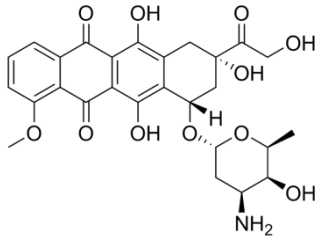
▶ Antitumor agents

: intercalators and chemical modifiers

1. Intercalators:



Daunorubicin



Doxorubicin

· small molecules that insert themselves into the structure of DNA (intercalation)

· anthracyclines:

– be derived from the genus *Streptomyces*
– ex) daunorubicin (daunomycin), doxorubicin (adriamycin), etc.
– <u>three mechanisms of action</u>:
  · inhibiting DNA and RNA synthesis by intercalating b/w base pairs of the DNA/RNA strand
  · creating free oxygen radicals → causing cellular damages (including DNA and cell membrane damages)
  · inhibiting topoisomerase II enzyme

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❖ Nucleic acid synthesis inhibitors

▶ Antitumor agents

: intercalators and chemical modifiers

1. Intercalators:

· small molecules that insert themselves into the structure of DNA (intercalation)

· anthracyclines:

– spectrum of activity:
  · some of the most effective anticancer treatments
  · effective against more types of cancer than any other anticancer agents

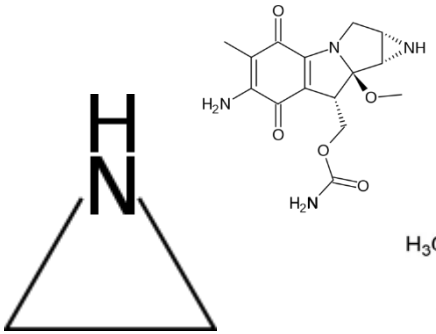
– side effects: cardiotoxicity, bone marrow suppression, hair loss, vomiting, etc.

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❖ Nucleic acid synthesis inhibitors

▶ Antitumor agents

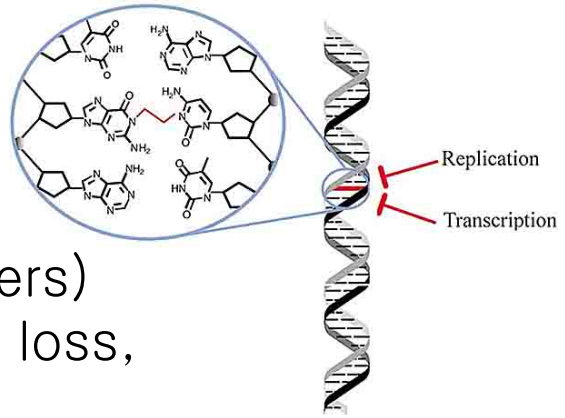
: intercalators and chemical modifiers



2. Chemical modifiers:

· mitomycin C:

- a family member of aziridine-containing natural product isolated from the genus *Streptomyces*
  · aziridine: three-membered heterocycle with one amine group and two methylene groups (C_2H_5N)
- <u>mechanism of action</u>: inducing DNA crosslinking
  · DNA crosslinking: linking of one polymer chain to another by alkylation (intra- and interstrand crosslinking)
- <u>spectrum of activity</u>: various solid tumors (e.g. gastrointestinal, anal, and breast cancers)
- <u>side effects</u>: bone marrow suppression, hair loss, lung fibrosis, renal damage, etc.



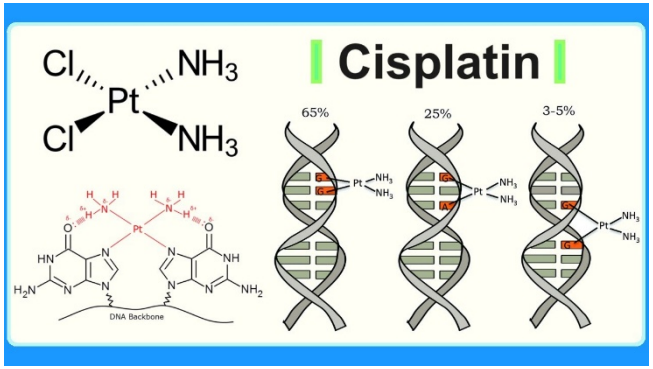
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❖ Nucleic acid synthesis inhibitors

▶ Antitumor agents

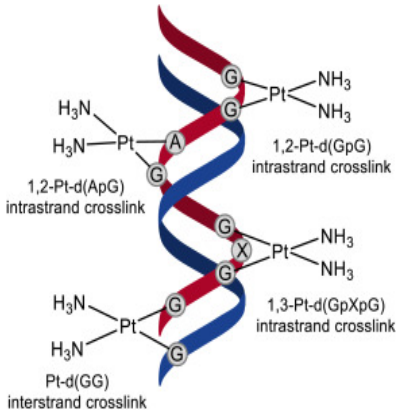
: intercalators and chemical modifiers

2. Chemical modifiers:



· cisplatin:

- a chemical compound (*cis*–$PtCl_2(NH_3)_2$)
- mechanism of action: DNA crosslinking and apoptosis
- spectrum of activity: various types of cancers (e.g. sarcoma, some carcinomas, lymphomas, etc.)
- side effects: bone marrow suppression, hair loss, nephrotoxicity, neurotoxicity, ototoxicity, nausea, vomiting, etc.



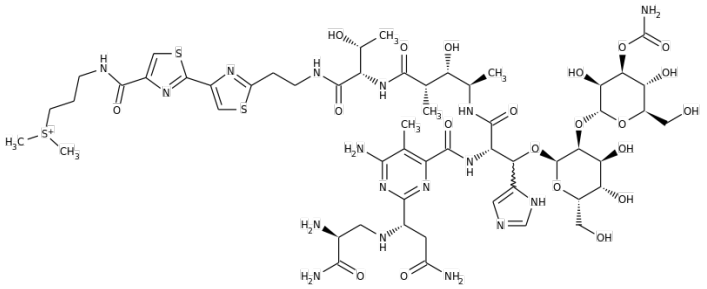
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❖ Nucleic acid synthesis inhibitors

▶ Antitumor agents

: intercalators and chemical modifiers

2. Chemical modifiers:



· **bleomycin**:

- be derived from the genus *Streptomyces* (*S. verticillus*)
- <u>mechanisms of action</u>:
  · acting by induction of DNA strand breaks
  (but, no known exact mechanism of DNA strand scission)
  · inhibiting the incorporation of thymidine into DNA strands
- <u>spectrum of activity</u>: skin cancers, Hodgkin's lymphoma, etc.
- <u>side effects</u>: pulmonary fibrosis / fever, rash, hair loss, discoloration of fingers and toes / minimal bone marrow suppression, etc.